

Scaling behaviors in simulated collagen fibrils

Robert Bertoldo Tavares

Departamento de Física, Universidade Federal do Ceará, Fortaleza, Ceará, Brasil

Michael Ferreira de Souza

*Departamento de Estatística e Matemática Aplicada,
Universidade Federal do Ceará, Fortaleza, Ceará, Brasil*

Béla Suki

*Department of Biomedical Engineering,
Boston University, Boston, MA, USA*

Ascânio Dias Araújo*

Departamento de Física, Universidade Federal do Ceará, Fortaleza, Ceará, Brasil and

*Departamento de Estatística e Matemática Aplicada,
Universidade Federal do Ceará, Fortaleza, Ceará, Brasil*

(Dated: March 18, 2026)

Abstract

Collagen fibrils are hierarchical protein assemblies that provide tensile strength and structural support to mammalian tissues. Understanding the relationship between assembly kinetics, structural organization, and mechanical properties is fundamental to soft matter physics, biomechanics and biomaterials science. Here, we investigate the formation of collagen fibrils using a modified diffusion-limited aggregation (DLA) model. Rod-like molecules attach to an existing core and carry out surface diffusion up a fixed number of step, T_s , to minimize the exposed surface area. Our findings indicate that increasing T_s results in a morphological transition from fractal aggregates to compact structures, characterized by a systematic evolution of the fibril's fractal dimension. We also develop a probabilistic fracture model to investigate the dependence of the fibril's mechanical failure on its architecture governed by T_s . Our results demonstrate that the rupture process evolves through stochastic avalanche events—a hallmark of Self-Organized Criticality (SOC). Specifically, the distribution P of avalanche sizes s follows a power-law scaling $P(s) \sim s^{-\gamma}$, indicating that the system lacks a characteristic scale. Furthermore, the scaling exponent γ is sensitive to the structural transition induced by T_s : γ increases from ≈ 2.3 at low T_s to ≈ 2.8 at high T_s , crossing the mean-field equal-load-sharing value $\gamma = 5/2$ and signaling a crossover from local to global load-sharing universality.

I. INTRODUCTION

Type I collagen is the principal structural protein of the mammalian extracellular matrix (ECM), providing essential mechanical integrity to tissues such as skin, tendon, bone, cartilage, and the cornea [1–7]. These hierarchical assemblies are critical for elasticity, tensile strength, and resilience against physical stress [1, 8]. Individual collagen molecules, characterized by a high aspect ratio ($\approx 300 \text{ nm} \times 1.5 \text{ nm}$) [1, 9], self-assemble into fibrils that can reach lengths of $500 \text{ }\mu\text{m}$ and diameters of 500 nm [10–12]. These fibrils further aggregate into fibers ($1\text{--}20 \text{ }\mu\text{m}$ diameter), forming a multiscale network capable of storing and transmitting mechanical energy [7, 13].

Characterizing the mechanical behavior of collagen fibrils, particularly at the nanoscale, presents significant challenges [14]. Experimental techniques such as atomic force microscopy

* ascanio@fisica.ufc.br

(AFM) and micro-mechanical actuation have successfully probed the tensile properties of isolated fibrils. Van der Rijt et al. used AFM to measure the mechanical properties of tendon fibrils, evaluating their tensile behavior until failure [15], while Svensson et al. [16] developed a piezoelectric actuator-based technique to measure stress and strain during uniaxial stretching. However, resolving the collective molecular dynamics that govern failure remains difficult due to experimental resolution limits, particularly when characterizing small-scale forces and intermolecular interactions [17]. Consequently, computational modeling has become indispensable for exploring the relationship between intrafibrillar organization and macroscopic mechanical response under a wider range of conditions.

Several theoretical frameworks have been developed to address fibril mechanics. Buehler et al. introduced coarse-grained molecular dynamics to model large-scale deformation, while Depalle et al. investigated the role of mineralization in stabilizing fibril elasticity in bone [18–22]. Other stochastic approaches have examined how enzymatic degradation alters tissue mechanics [23] and how morphological features like waviness influence fiber modulus [24]. Despite these advances, the role fractal organization may play in the mechanical resilience and the statistical dynamics of rupture remains largely unexplored.

In this work, we investigate the interplay between fractal structure and failure mechanics of collagen fibrils. We employ a modified Diffusion-Limited Aggregation (DLA) model incorporating a surface diffusion process, controlled by the number of diffusion steps T_s that regulates the degree of lateral compaction and minimizes the exposed surface area [25]. By coupling this structural model with a probabilistic fracture simulation, we show that T_s governs both the fractal dimension and the mechanical strength of the fibril. Although the fracture model is inherently stochastic, it captures the essential role that structural compaction plays in determining the fibril’s mechanical response. This model reveals that the rupture process exhibits avalanche dynamics consistent with power-law scaling distributions characteristic of Self-Organized Criticality (SOC).

II. MODEL AND RESULTS

A. Fibril Formation Model

We modeled collagen fibril formation using a three-dimensional diffusion-limited aggregation (DLA) algorithm on a cubic lattice. Collagen molecules were represented as rigid rods (rectangular bars) with dimensions $L_x = 1$, $L_y = 18$, and $L_z = 1$ in lattice units (l.u.), with L_y defining the characteristic length scale of the model.

The aggregation begins by placing the first molecule at the lattice center. Subsequent molecules are released one at a time from random positions on a sphere of radius R centered at the origin. The radius R is updated to track the maximum extent of the growing aggregate along the y -axis. Each molecule then performs a random walk on the lattice until one of the following conditions is met:

1. The molecule attaches to the existing aggregate;
2. The molecule is discarded if its distance from the origin exceeds $2R$.

After each attachment, R is recalculated. For simplicity, molecular rotation during diffusion is not allowed. This cycle of release, diffusion, and attachment continues until the aggregate reaches the desired size, limited here to 3×10^4 molecules.

Diffusion is implemented as a three-dimensional random walk on the cubic lattice. In the x - z plane, molecules can move to first-neighbor sites (1 l.u.) or second-neighbor sites ($\sqrt{2}$ l.u., i.e., diagonal moves). Along the y -axis, movement is restricted to first-neighbor sites (1 l.u.). In real collagen fibrillogenesis, lateral aggregation occurs with a characteristic axial spacing of 67 nm [11], which constrains the positions where new molecules can be incorporated. In our model, molecular attachment is restricted to multiples of 4 l.u. along the y -axis [25].

The formation of real fibrils is driven by electrostatic forces [25] that favor lateral association and minimize exposed surface area [25, 26]. To mimic this relaxation, we implemented a lateral surface-diffusion step after attachment: a newly incorporated molecule performs a random walk restricted to the x - z plane (no movement along the y -axis), using the same rules described above. The molecule is allowed up to T_s diffusion attempts to search the local surface and is placed at the position that minimizes its exposed area; if multiple positions

are equivalent, the first one encountered is retained [27]. Using this algorithm, we generated ensembles of fibrils for different values of T_s to examine how post-attachment mobility controls fibril compactness, structure, and subsequent mechanical response.

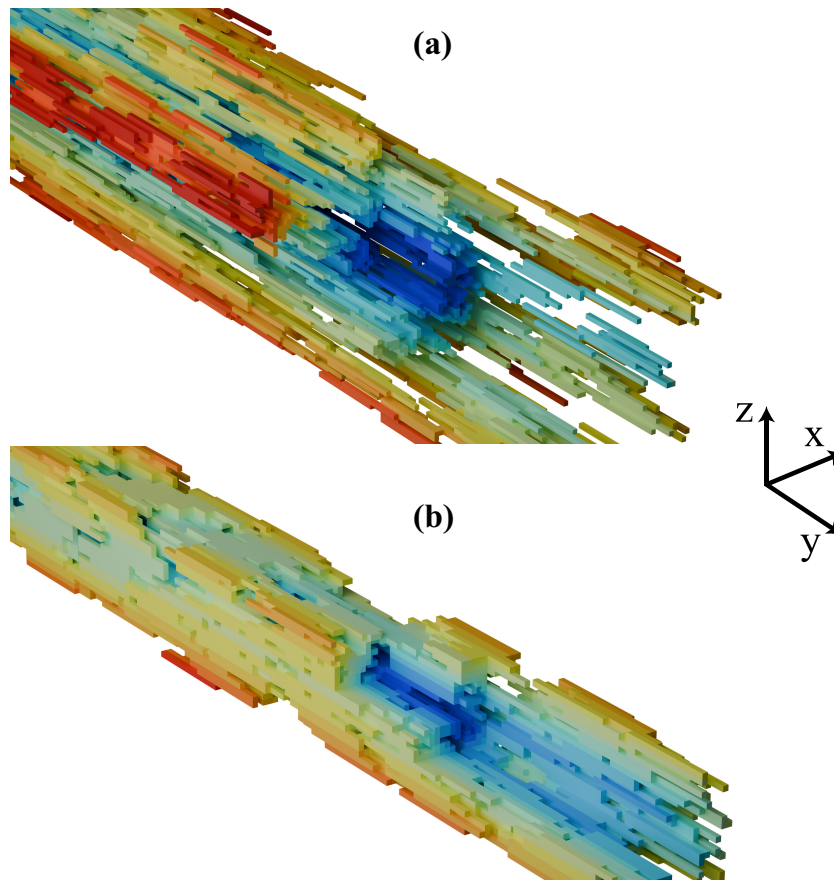


Figure 1. Representative fibril segments generated with the DLA algorithm containing 30,000 molecules. Panels (a) and (b) show typical fibrils obtained with $T_s = 2$ (short-term diffusion) and $T_s = 8192$ (long-term diffusion), respectively. Colors indicate the relative time of attachment, from early (blue) to late (red).

The generated fibrils exhibit complex, heterogeneous packing with internal voids whose prevalence depends strongly on T_s (Fig. 1). For $T_s = 2$, the fibril displays an open architecture with visible gaps between molecular clusters. In contrast, for $T_s = 8192$, the structure becomes more compact and homogeneous, with substantially reduced void space.

Although fibril growth proceeds primarily along the longitudinal axis (y -direction), the influence of the surface diffusion parameter T_s is most pronounced in the radial organization of the structure (x - z plane), as illustrated in Fig. 2. At low T_s (Fig. 2a), the projected

91 cross-sections are open and branched. As T_s increases, however, the packing becomes pro-
 92 gressively more compact and regular, reaching a highly uniform configuration at $T_s = 8192$
 93 (Fig. 2d). Since a higher local packing density implies a greater number of intermolecu-
 94 lar contacts, this morphological transition suggests that fibrils formed at larger T_s exhibit
 95 enhanced connectivity and mechanical stability [28].

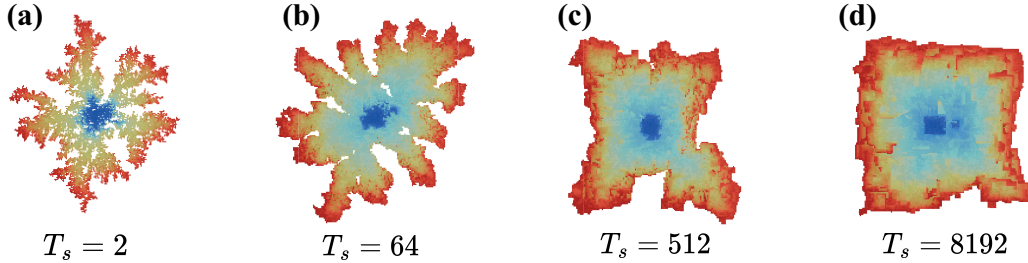


Figure 2. Representative examples of projected central segments (x - z plane) for different values of T_s : (a) $T_s = 2$, (b) $T_s = 64$, (c) $T_s = 512$, and (d) $T_s = 8192$. The progression illustrates a clear transition from sparse, irregular morphology at low T_s to dense, radially symmetric packing at high T_s . The color gradient (blue to red) represents the temporal sequence of molecular incorporation, from early-attached to recently added molecules.

96 To quantify how T_s modulates fibril structure, we estimated the fractal dimension of the
 97 central segment, defined as the region extending from $y = -100$ to $y = 100$. For each T_s , we
 98 generated 50 fibrils and isolated this central region from each. Within each central segment,
 99 we extracted 11 cross-sections at intervals of 18 l.u. between $y = -90$ and $y = 90$, yielding
 100 a total of 550 cross-sections per T_s value. For each cross-section, we computed the center
 101 of mass and the maximum radius enclosing all particles (segments of molecules); $R_{\max}$ was
 102 subsequently defined as the largest of these 550 radii. The mass $m(R)$ was measured as the
 103 number of particles within a circular region of radius R centered at the cross-section's center
 104 of mass. For each value of R spanning from $R_{\min} = 5$ l.u. to $R_{\max}$, we computed the mean
 105 mass $\langle m(R) \rangle$ by averaging over all 550 cross-sections [29]. The fractal dimension D_f was
 106 then estimated from the mass-radius scaling relation [30]:

$$\langle m(R) \rangle \sim R^{D_f}, \quad (1)$$

107 by performing a linear fit on a log-log plot of $\langle m(R) \rangle$ versus R .

As shown in Fig. 3, D_f depends strongly on T_s , providing a quantitative measure of the morphological transition observed in Fig. 2. For $T_s = 2$, we find that $D_f = 1.708 \pm 0.005$, close to the characteristic value of two-dimensional diffusion-limited aggregation ($D_f \approx 1.71$) [31]. As T_s increases, D_f rises and saturates at $D_f = 1.963 \pm 0.001$ for $T_s = 8192$, approaching the Euclidean limit of $D_f = 2.0$ expected for a compact two-dimensional disk. Thus, increasing post-attachment mobility drives a crossover from diffusion-limited, fractal packing to nearly space-filling cross-sections.

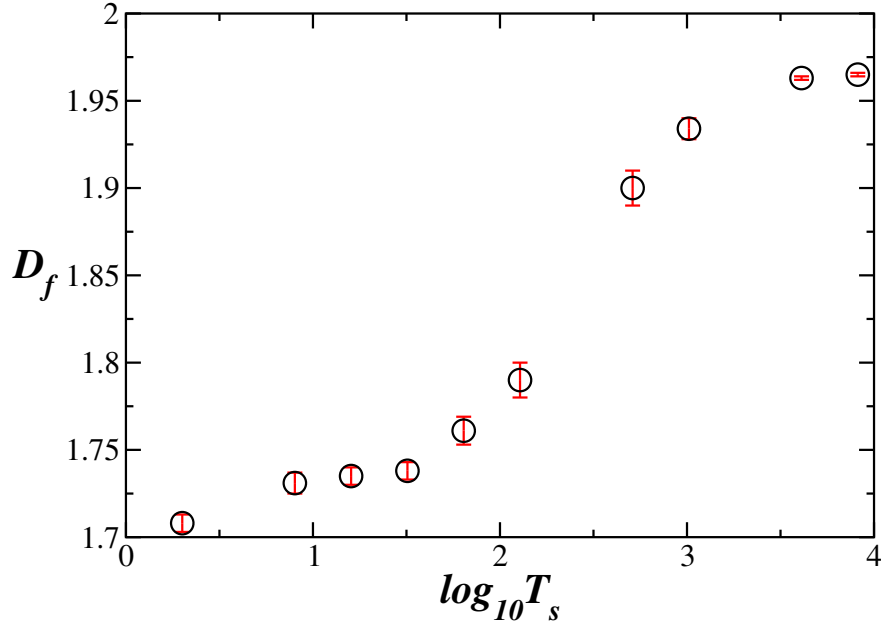


Figure 3. Average fractal dimension D_f as a function of the diffusion parameter T_s on a log-linear plot. Error bars are shown in red. The fractal dimension increases from $D_f = 1.708 \pm 0.005$ at $T_s = 2$ to $D_f = 1.963 \pm 0.001$ for $T_s = 8192$.

B. Mechanical Model for Fibril Rupture

To connect the T_s -dependent fibril architecture to mechanical failure, we modeled rupture using a progressive, probabilistic fracture framework commonly employed for disordered networks [32–34]. Because collagen molecules are represented as rigid rods that do not undergo continuous deformation prior to failure, we do not attempt to resolve deterministic crack propagation. Instead, intermolecular bonds fail stochastically under an applied axial force, and whole molecules detach from the load-bearing network with a stress-dependent

probability. The procedure consists of: (i) isolating a trunk (force-carrying core) with dimensions $S_x = 17$, $S_y = 201$, and $S_z = 17$ l.u. [35]. Truncation inevitably creates disconnected dangling ends that do not contribute to force transmission [36]; we therefore (ii) identify the backbone (active skeleton) within the trunk prior to applying tensile loading [37].

Backbone extraction proceeds in two passes. In the first pass, molecules at $y = 1$ are marked as active, and we scan upward to $y = S_y$, progressively marking molecules in contact with active ones; unmarked molecules are then removed. In the second pass, molecules at $y = S_y$ are marked as active, and we scan downward to $y = 1$ using the same connectivity rule; only molecules marked active after this pass are retained, defining the final backbone [35] (Fig. 4).

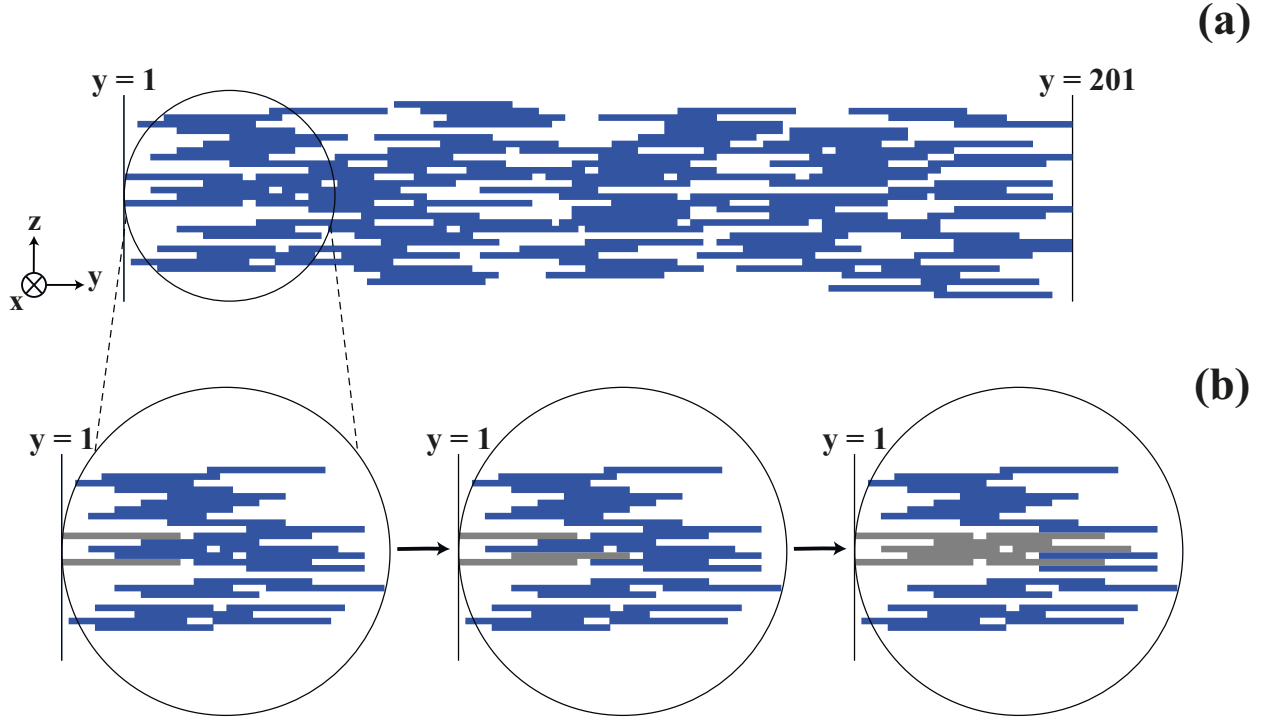


Figure 4. (a) Two-dimensional visualization of an isolated fibril trunk used to extract the backbone. The circle highlights the region where the identification procedure starts ($y = 1$). (b) Example of the backbone identification procedure from left to right: molecules marked as active are highlighted in grey as connectivity is propagated along the trunk, yielding the final backbone.

With the backbone identified, we simulate the effects of axial tension along the fibril's axis (y) by progressively increasing a dimensionless force parameter F , where F regulates

the removal of molecules in a probabilistic fashion. At each cross-section i , we define the effective load-bearing area as the number of backbone segments intersecting that plane, $N(i)$. Each discrete segment is assigned a unit area on the lattice [25]; assuming uniform load sharing, the local tensile stress at cross-section i is:

$$\sigma(i) = \frac{F}{N(i)}. \quad (2)$$

We assume that local failure occurs by rupturing intermolecular bonds so that molecules detach from each other rather than break [35]. For a molecule spanning n cross-sections, the mean stress is computed as the average of the local cross-sectional stresses:

$$\sigma_M = \frac{1}{n} \sum_{i=1}^n \sigma(i). \quad (3)$$

A molecule's resistance to detach is assumed to scale with the number of intermolecular contacts it forms within the backbone [38]. For each molecule, we count first-neighbor contacts for every segment in each intersected cross-section and sum these contributions to obtain the total coordination number K . The corresponding failure threshold is $K\sigma_c$, where σ_c is the critical stress for bond rupture; we set $\sigma_c = 1$, assuming identical molecules. To simplify the model, both the applied force parameter F and the critical stress σ_c are treated as dimensionless quantities in arbitrary units. Hence, $\sigma_c = 1$ represents the normalized threshold for intermolecular bond rupture, which allows us to focus on relative mechanical behavior rather than absolute physical values.

The probability of removing a molecule is then given by:

$$P_R = \left(\frac{\sigma_M}{K\sigma_c} \right)^m, \quad (4)$$

where m is the Weibull modulus (dimensionless shape parameter) that controls the amount of statistical scatter in this stochastic removal rule [35, 39]. Larger m yields a narrower distribution of effective rupture strengths (less variability), whereas smaller m produces broader scatter. Here, we set $m = 2$ following the work of Parkinson et al. [35]. Fig. 5 schematically illustrates the calculation of $\sigma(i)$, σ_M , and K for a representative molecule.

For a given force F , we evaluate P_R for every molecule in the backbone and compare it with an independent random number $u \in [0, 1]$. If $u < P_R$, the molecule is removed. After each sweep, stresses and probabilities are recalculated for the remaining backbone, and the

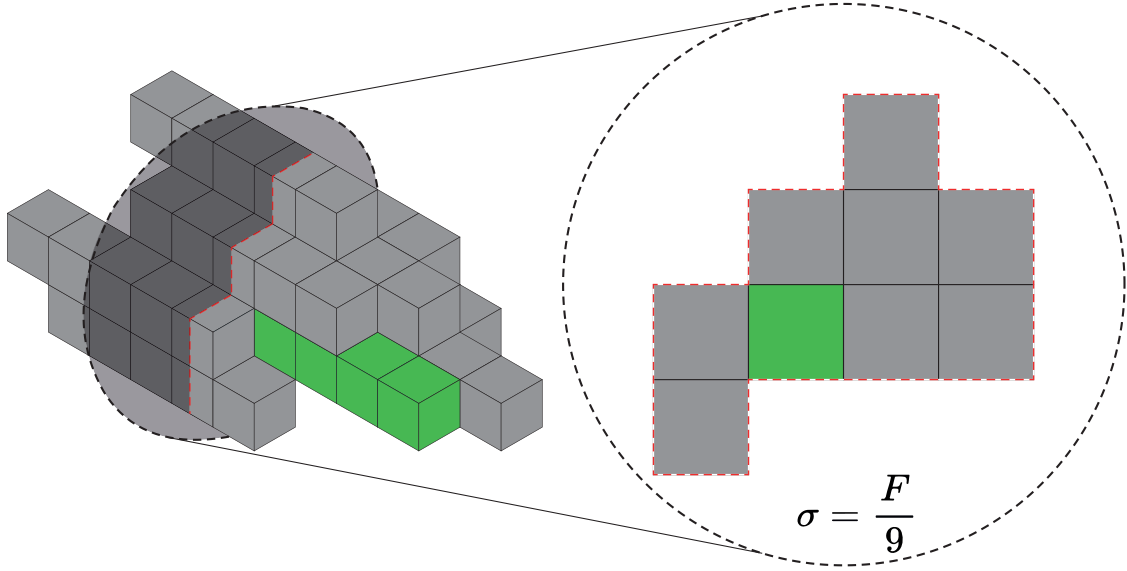


Figure 5. Schematic illustration of the computation of the removal probability P_R for a representative molecule (green). Left: a portion of the backbone where each rectangle represents a molecule (drawn with five segments for clarity). The green molecule intersects multiple cross-sections; for each intersected cross-section i , the local stress is $\sigma(i) = F/N(i)$ (right; example with $N(i) = 9$, giving $\sigma(i) = F/9$).

160 sweep is repeated until no additional removals occur. The applied force is then increased
 161 by $\Delta F = 0.5$, and the procedure is repeated. Rupture is defined as the appearance of a
 162 completely void cross-section, signaling failure of the backbone. The steps of this removal
 163 process are schematized in Fig. 6.

164 Since our rupture model is probabilistic, for each value of T_s we analyzed an ensemble of
 165 10 distinct fibrils, each containing 3×10^4 molecules, and performed 10^3 independent rupture
 166 simulations per fibril. During each simulation, the force was increased gradually, and at each
 167 level we recorded the molecules removed from the backbone.

168 To quantify the mechanical resistance of the fibrils, we analyzed the evolution of the
 169 average fraction of removed molecules, denoted by φ , as a function of the applied force
 170 F [35]. As shown in Fig. 7(a), φ increases monotonically with F , reflecting progressive
 171 intermolecular bond ruptures and subsequent loss of load-bearing connectivity.

172 The force threshold for complete fibril rupture increases with T_s , consistent with enhanced
 173 connectivity in more compact fibrils. Despite different rupture thresholds, all curves exhibit
 174 a common functional form:

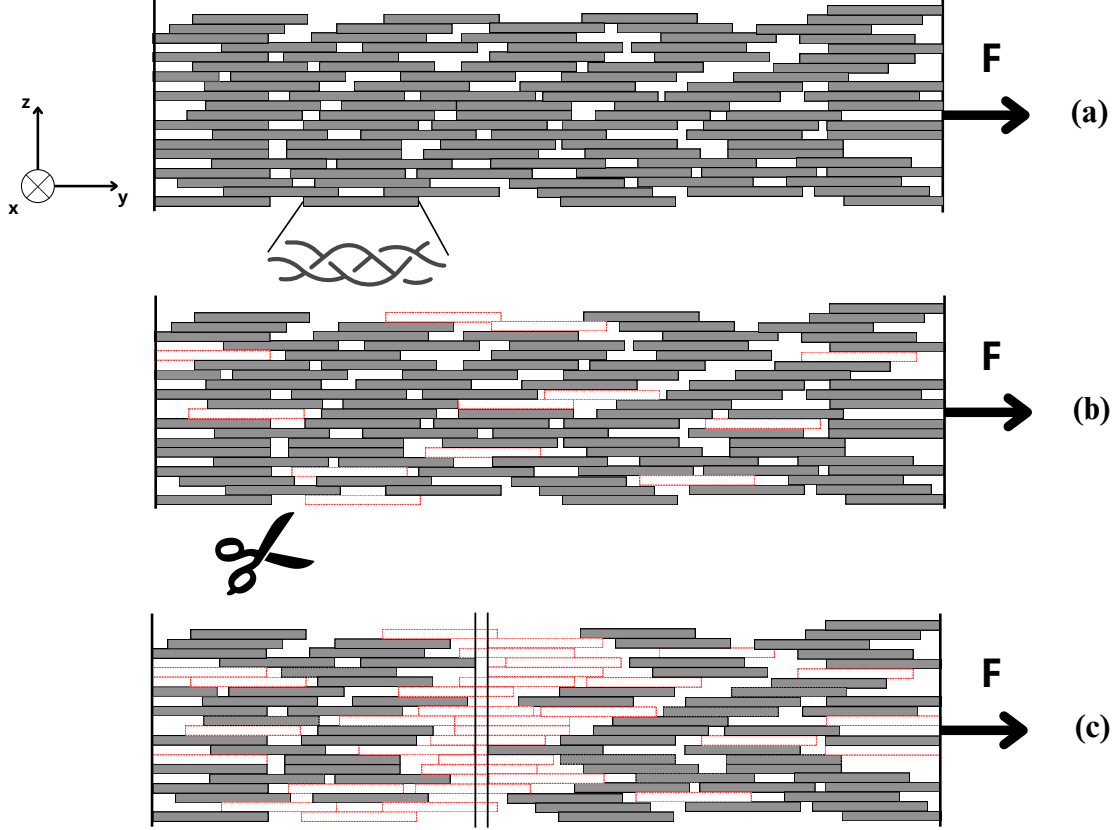


Figure 6. Schematic view of the load-bearing backbone and the stages of the rupture process. (a) Two-dimensional view of the backbone subjected to an axial force F along the y -axis. Each molecule (gray) is assigned a removal probability P_R (Eq. 4); a molecule is removed when a random number $u \in [0, 1]$ satisfies $u < P_R$. (b) Progressive damage after some molecules have been removed, represented by red dashed lines. (c) Rupture occurs when a cross-section becomes completely empty (solid lines), eliminating the continuous load path.

$$f(F) = 10^{-3} [\exp(\beta F) - 1 + F^\alpha], \quad (5)$$

where the coefficients α and β are determined from fits to the data by using Eq. (5). At low forces, the response is dominated by the power-law term, corresponding to gradual damage accumulation. Close to rupture, the exponential term dominates, producing a rapid increase in φ [40]. Fig. 7(b) and Fig. 7(c) show that both α and β decrease with T_s and stabilize for $T_s \geq 512$. The parameter α sets the rate of gradual (power-law) damage accumulation [41], whereas β controls the onset of rapid, avalanche-driven failure [42, 43].

During rupture, molecules detach either individually or in spatially connected clusters

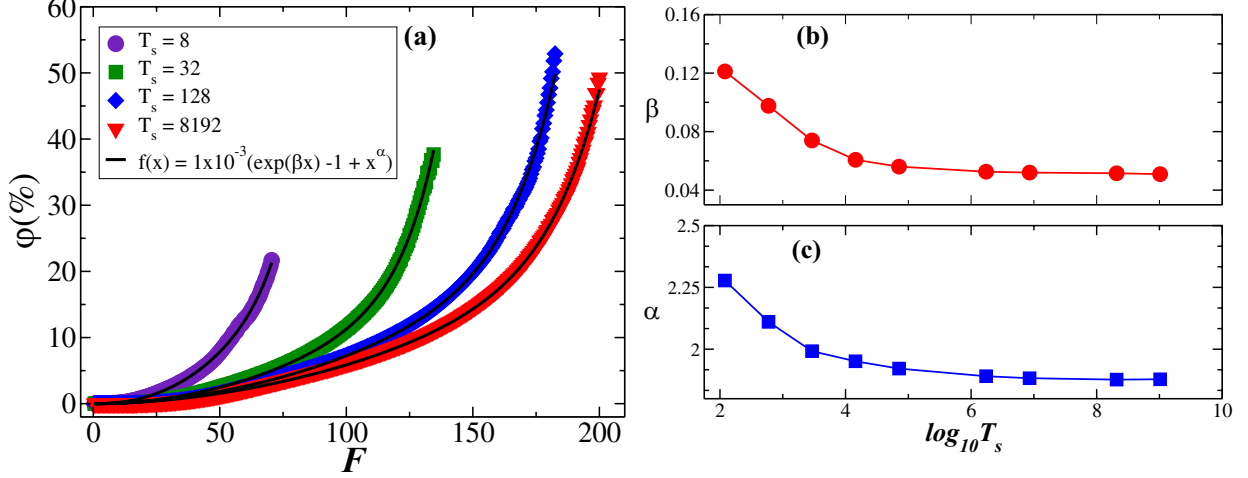


Figure 7. (a) Average fraction of removed molecules φ versus applied force F for $T_s = 8, 32, 128$, and 8192. Solid lines are fits to $f(F) = 10^{-3}[\exp(\beta F) - 1 + F^\alpha]$. (b) and (c) Fitted parameters β and α as a function of T_s ; both decrease reaching a constant for $T_s \geq 512$.

removed simultaneously during the same force increment. We define a cluster as any group of two or more adjacent molecules removed at the same force step. A cluster of such detachments corresponds to an avalanche, a hallmark of crackling dynamics in driven disordered systems [44–49]. To quantify the importance of collective events, we introduce Ψ , defined as the fraction of removed molecules that belong to clusters relative to the total number removed molecules at a given force. To compare across different rupture forces, we express the force as a normalized variable $F_n = F/F_{\text{rup}}$, where F_{rup} is the rupture force of each realization. As shown in Fig. 8, low- T_s fibrils exhibit large Ψ even at small F_n , indicating early collective failure, whereas high- T_s fibrils primarily fail through isolated removals until close to rupture, when Ψ rises sharply.

To understand this collective behavior, we analyze it from the viewpoint of avalanche statistics [50]. Here, we define the avalanche size, s , as the cluster size, i.e., the number of molecules in a removed cluster at a fixed force level. We then compute $P(s)$, the probability distribution of avalanche sizes, for each value of T_s . The inset of Fig. 9(a) shows representative distributions for $T_s = 2$ and $T_s = 8192$. To characterize these distributions, we apply the scaling law [51]:

$$P(s) \sim s^{-\gamma}, \quad (6)$$

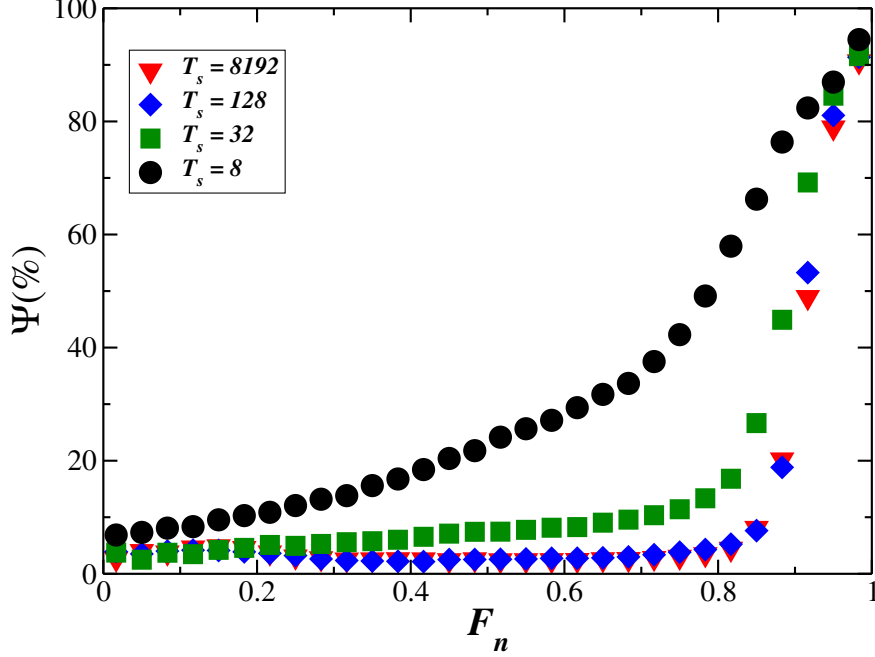


Figure 8. The average fraction of molecules removed in a cluster, Ψ , as a function of normalized force $F_n = F/F_{\text{rup}}$ for $T_s = 8, 32, 128$ and 8192 . For low T_s , local failure due to molecules detaching in clusters contributes significantly even at low forces, indicating early onset of collective failure. In contrast, for high T_s , damage accumulation in the fibril initially is driven by isolated bond breakings, whereas rupture in cluster (and catastrophic failure) occurs only near full fibril rupture.

where γ is the exponent of the power law.

The main panel in Fig. 9(a) shows the scaling regime where this power-law behavior provides a good fit over two orders of magnitude. Our analysis yields $\gamma = 2.31 \pm 0.05$ for $T_s = 2$ and $\gamma = 2.80 \pm 0.04$ for $T_s = 8192$. Between these limits, γ varies systematically with T_s (Fig. 9b): it increases approximately linearly with $\log(T_s)$ for $T_s < 512$ and reaches a plateau for $T_s \geq 512$.

In many stochastic fracture and fiber-bundle models, the avalanche exponent γ is sensitive to how stress is redistributed following a local failure event, with a crossover from local to global load sharing [43, 52–57]. In our fibrils, increasing T_s drives a structural transition toward compact packing (larger D_f , Fig. 3), implying a more densely connected backbone and hence a larger effective stiffness κ_{eff} . Since Eq. (1) implies that in a $d = 2$ cross-section $\rho(R) \sim \langle m(R) \rangle / R^2 \sim R^{D_f-2}$, the approach $D_f \rightarrow 2$ corresponds to an increasingly space-filling morphology with weak scale dependence of ρ . A larger κ_{eff} enables broader stress

redistribution, de-correlates rupture events, and suppresses system-spanning avalanches, making small failure events more frequent relative to large ones and thereby increasing γ [50, 58–62]. Consistent with this interpretation, the saturation of γ for $T_s \geq 512$ coincides with $D_f \gtrsim 1.90$ and with the stabilization of the fit parameters α and β (Fig. 7), suggesting diminishing sensitivity of the failure statistics to further surface relaxation. Furthermore, the systematic increase in γ with T_s signals a fundamental shift in the system’s criticality: the transition from fractal to compact architecture progressively screens long-range elastic interactions, driving the failure dynamics toward a regime dominated by small-scale stochastic events and thus pointing to a change in universality class.

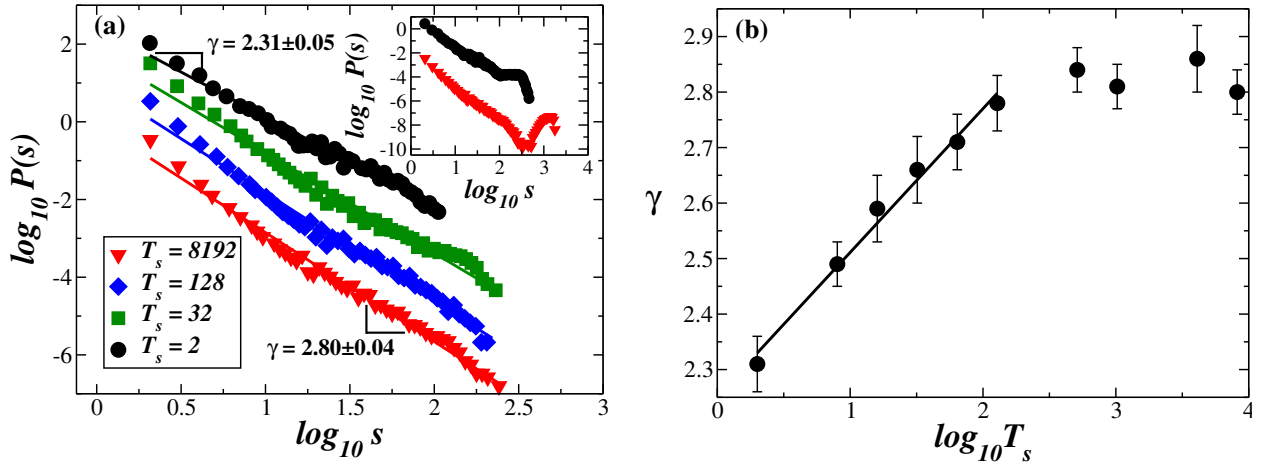


Figure 9. (a) Log-log plot of the rupture avalanche size distribution, $P(s)$, as a function of avalanche size s for $T_s = 2, 32, 128$ and 8192 . A power-law regime $P(s) \sim s^{-\gamma}$ is observed, with slopes γ ranging from 2.31 ± 0.04 ($T_s = 2$) to 2.80 ± 0.05 ($T_s = 8192$) obtained from the linear fit (solid lines) following the scaling behavior. The inset shows the full distributions, including the tails, for $T_s = 2$ and $T_s = 8192$. The avalanche size distributions are vertically shifted for better visualization. In (b), the exponent γ is plotted as a function of T_s on a log-linear scale, showing two regimes: growth for $T_s < 512$ and a plateau for $T_s \geq 512$.

A direct consequence of avalanche-like rupture is that sudden stress drops should exist in the stress–strain curve, since avalanches propagate faster than the imposed loading rate [34]. Although our model does not predict strain and thus cannot generate stress–strain curves, experimental evidence strongly supports this prediction. Gutsman et al. [63] used force spectroscopy on single microfibrils and directly observed force drops caused by intermolecular cross-link ruptures. Svensson et al. [64] found clear stress drops in tensile tests of individual

collagen fibrils, while Deng et al. [24] observed similar drops in the stress–strain curves.

Our coarse-grained framework represents collagen molecules as rigid rods with restricted movement and no rotational diffusion. Intermolecular interactions are simplified via T_s , which lacks a direct mapping to interaction changes induced by biochemical conditions such as pH [65]. Furthermore, while DLA captures the stochastic character of molecular accretion, biological fibrillogenesis is an actively regulated, energy-dependent process: collagen molecules first self-assemble into microfibrils—typically containing five molecules—which are then actively positioned and concatenated by the cell to form the growing fibril [11]. Fracture is implemented as a probabilistic model, omitting detailed force transmission and internal bond-breaking mechanics [66, 67]. Consequently, the model does not produce stress–strain curves or elastic moduli [68]. Nevertheless, it reveals that rupture follows scale-free avalanche dynamics and that structural compactness correlates with mechanical resilience. Thus, the fibril’s fractal dimension serves as a quantitative bridge between structural compactness and failure statistics.

III. CONCLUSION

In this work, we combined a three-dimensional diffusion-limited aggregation model of collagen fibrillogenesis with a progressive probabilistic rupture model to connect assembly kinetics, fibril architecture, and failure statistics. Increasing the post-attachment surface diffusion parameter T_s drives a clear morphological crossover from porous, fractal packing to nearly compact cross-sections: the measured fractal dimension rises from $D_f = 1.708 \pm 0.005$ at $T_s = 2$ to $D_f = 1.963 \pm 0.001$ at $T_s = 8192$. Mechanically, more compact fibrils exhibit higher rupture thresholds and a delayed onset of collective failure, while the avalanche size distribution remains scale-free, $P(s) \sim s^{-\gamma}$. Importantly, the avalanche exponent increases systematically with T_s , from $\gamma = 2.31 \pm 0.05$ ($T_s = 2$) to $\gamma = 2.80 \pm 0.04$ ($T_s = 8192$), with a plateau for $T_s \geq 512$ that coincides with $D_f \gtrsim 1.90$ and with saturation of the damage-related parameters α and β .

From a physical perspective, the joint increase of D_f and γ indicates that increasing the density of the load-bearing backbone promotes more homogeneous stress redistribution, shifting the rupture dynamics away from correlated, system-spanning avalanches and toward a regime dominated by smaller events, consistent with a crossover in the effective universality

256 class.

257 In the framework of fiber-bundle models, the exponent $\gamma = 5/2$ is the exact mean-field
258 result for equal load sharing (ELS) [53–55], whereas local load sharing (LLS) yields lower,
259 non-universal exponents that depend on dimensionality and disorder [58–61]. Our measured
260 exponents evolve from $\gamma \approx 2.3$ at low T_s —below the ELS value and thus closer to the
261 LLS regime—to $\gamma \approx 2.8$ at high T_s , exceeding $5/2$ and approaching the behavior expected
262 when stress redistribution becomes effectively global [57]. This progression suggests that
263 increasing structural compactness extends the range of load redistribution within the fibril,
264 driving a genuine transition from local to global load-sharing universality.

265 From a biological standpoint, the model developed here provides a simple mechanism
266 to tune mechanical resilience of collagen fibrils, fibers and tissues by controlling assembly
267 conditions that regulate compactness. We speculate that compactness of collagen fibrils
268 depend on evolutionarily fine-tuned macroscopic function such as the load the fibrils are
269 required to carry in a given tissue or organ. In diseases, microstructural remodeling of fibrils
270 alters their compactness that can compromise their macroscopic robustness leading to fibril
271 failure as in pulmonary emphysema, aortic dissection and aneurysm. Future work should
272 incorporate more realistic mechanics and time dependence—including elastic or viscoelastic
273 deformation and evolving stiffness (e.g., due to cross-link dynamics or degradation)—to
274 predict stress–strain behavior and track how the fiber modulus changes over time under
275 load.

276 IV. AUTHOR CONTRIBUTIONS

277 All authors contributed equally to this work.

278 V. ACKNOWLEDGMENTS

279 We thank the Brazilian agencies CNPq, CAPES, and FUNCAP for financial support.

280 [1] F. H. Silver, M. Jaffe, and R. G. Shah, in *Handbook of Properties of Textile and Technical*
281 *Fibres* (Elsevier, 2018) pp. 345–365.

- 282 [2] E. Davison-Kotler, W. S. Marshall, and E. García-Gareta, *Bioengineering (Basel)* **6**, 56 (2019).
- 283 [3] I. Tresoldi, F. Oliva, M. Benvenuto, M. Fantini, L. Masuelli, R. Bei, and A. Modesti, *Muscles*
284 *Ligaments Tendons J.* **3**, 2 (2013).
- 285 [4] N. S. Fedarko, in *Osteogenesis Imperfecta* (Elsevier, 2014) pp. 45–56.
- 286 [5] G. A. Rico-Llanos, S. Borrego-González, M. Moncayo-Donoso, J. Becerra, and R. Visser,
287 *Polymers (Basel)* **13**, 599 (2021).
- 288 [6] S. Chen, M. J. Mienaltowski, and D. E. Birk, *Exp. Eye Res.* **133**, 69 (2015).
- 289 [7] B. Suki, *Structure and function of the extracellular matrix* (Academic Press, San Diego, CA,
290 2021).
- 291 [8] I. N. Amirrah, Y. Lokanathan, I. Zulkiflee, M. F. M. R. Wee, A. Motta, and M. B. Fauzi,
292 *Biomedicines* **10**, 2307 (2022).
- 293 [9] K. Gelse, E. Pöschl, and T. Aigner, *Adv. Drug Deliv. Rev.* **55**, 1531 (2003).
- 294 [10] J. Charvolin and J.-F. Sadoc, *Eur. Phys. J. E Soft Matter* **42**, 49 (2019).
- 295 [11] K. E. Kadler, D. F. Holmes, J. A. Trotter, and J. A. Chapman, *Biochem. J.* **316**, 1 (1996).
- 296 [12] D. A. D. Parry and A. S. Craig, in *Ultrastructure of the Connective Tissue Matrix* (Springer
297 US, Boston, MA, 1984) pp. 34–64.
- 298 [13] F. H. Silver and W. J. Landis, in *Collagen* (Springer US, Boston, MA, 2008) pp. 133–154.
- 299 [14] M. Nalbach, F. Chalupa-Gantner, F. Spoerl, V. de Bar, B. Baumgartner, O. G. Andriotis,
300 S. Ito, A. Ovsianikov, G. Schitter, and P. J. Thurner, *Rev. Sci. Instrum.* **93**, 054103 (2022).
- 301 [15] J. A. J. van der Rijt, K. O. van der Werf, M. L. Bennink, P. J. Dijkstra, and J. Feijen,
302 *Macromol. Biosci.* **6**, 697 (2006).
- 303 [16] R. B. Svensson, S. T. Smith, P. J. Moyer, and S. P. Magnusson, *Acta Biomater.* **70**, 270
304 (2018).
- 305 [17] O. G. Andriotis, M. Nalbach, and P. J. Thurner, *Acta Biomater.* **163**, 35 (2023).
- 306 [18] M. J. Buehler, *Proc. Natl. Acad. Sci. U. S. A.* **103**, 12285 (2006).
- 307 [19] M. J. Buehler, *J. Mater. Res.* **21**, 1947 (2006).
- 308 [20] M. J. Buehler, *Curr. Appl. Phys.* **8**, 440 (2008).
- 309 [21] M. J. Buehler, *J. Mech. Behav. Biomed. Mater.* **1**, 59 (2008).
- 310 [22] B. Depalle, Z. Qin, S. J. Shefelbine, and M. J. Buehler, *J. Bone Miner. Res.* **31**, 380 (2016).
- 311 [23] A. D. Araújo, A. Majumdar, H. Parameswaran, E. Yi, J. L. Spencer, M. A. Nugent, and
312 B. Suki, *Proc. Natl. Acad. Sci. U. S. A.* **108**, 9414 (2011).

- [24] Y. Deng, J. Herrmann, Y. Wang, M. Nguyen, J. K. Hall, J. H. Kim, M. L. Smith, K. R. Lutchén, E. Bartolák-Suki, and B. Suki, *Sci. Rep.* **14**, 29008 (2024).
- [25] J. Parkinson, K. E. Kadler, and A. Brass, *J. Mol. Biol.* **247**, 823 (1995).
- [26] K. E. Kadler, Y. Hojima, and D. J. Prockop, *J. Biol. Chem.* **262**, 15696 (1987).
- [27] J. M. García-Ruiz and F. Otálora, *Physica A* **178**, 415 (1991).
- [28] M. Mohammadkhah and S. Klinge, *J. Mech. Behav. Biomed. Mater.* **148**, 106203 (2023).
- [29] T. Vicsek, *Fractal growth phenomena (2nd edition)*, 2nd ed. (World Scientific Publishing, Singapore, Singapore, 1991).
- [30] N. J. Giordano, *Computational physics* (Pearson Education India, 2012).
- [31] T. A. Witten and L. M. Sander, *Phys. Rev. B Condens. Matter* **27**, 5686 (1983).
- [32] Z. P. Bažant and J. Planas, *Fracture and size effect in concrete and other quasibrittle materials* (Routledge, 2019).
- [33] A. Gilabert, C. Vanneste, D. Sornette, and E. Guyon, *Journal Phys.* **48**, 763 (1987).
- [34] H. Noguchi and S. Yukawa, *Phys. Rev. E* **110**, 045001 (2024).
- [35] J. Parkinson, A. Brass, G. Canova, and Y. Brechet, *J. Biomech.* **30**, 549 (1997).
- [36] H. J. Herrmann, D. C. Hong, and H. E. Stanley, *J. Phys. A Math. Gen.* **17**, L261 (1984).
- [37] A. A. Moreira, C. L. N. Oliveira, A. Hansen, N. A. M. Araújo, H. J. Herrmann, and J. S. Andrade, Jr, *Phys. Rev. Lett.* **109**, 255701 (2012).
- [38] C. A. Vater, E. D. Harris, Jr, and R. C. Siegel, *Biochem. J.* **181**, 639 (1979).
- [39] D. R. Jones and M. F. Ashby, *Engineering materials 2: an introduction to microstructures and processing* (Butterworth-Heinemann, 2012).
- [40] M. J. Buehler and Y. C. Yung, *Nat. Mater.* **8**, 175 (2009).
- [41] S. P. Veres, J. M. Harrison, and J. M. Lee, *J. Orthop. Res.* **31**, 731 (2013).
- [42] S. Zapperi, P. Ray, H. E. Stanley, and A. Vespignani, *Phys. Rev. Lett.* **78**, 1408 (1997).
- [43] S. Zapperi, P. Ray, H. E. Stanley, and A. Vespignani, *Phys. Rev. E Stat. Phys. Plasmas Fluids Relat. Interdiscip. Topics* **59**, 5049 (1999).
- [44] H. M. Jaeger and S. R. Nagel, *Science* **255**, 1523 (1992).
- [45] C. Godano, M. L. Alonzo, and V. Caruso, *Phys. Earth Planet. Inter.* **80**, 117 (1993).
- [46] J. M. Beggs and D. Plenz, *J. Neurosci.* **23**, 11167 (2003).
- [47] A. L. Barabási, S. V. Buldyrev, H. E. Stanley, and B. Suki, *Phys. Rev. Lett.* **76**, 2192 (1996).

- [48] A. M. Alencar, S. V. Buldyrev, A. Majumdar, H. E. Stanley, and B. Suki, Phys. Rev. Lett. **87**, 088101 (2001).
- [49] B. Suki, A. L. Barabási, Z. Hantos, F. Peták, and H. E. Stanley, Nature **368**, 615 (1994).
- [50] P. Bak, C. Tang, and K. Wiesenfeld, Phys. Rev. Lett. **59**, 381 (1987).
- [51] S. Zapperi, A. Vespignani, and H. E. Stanley, Nature **388**, 658 (1997).
- [52] H. E. Daniels, Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences **183**, 405 (1945).
- [53] P. C. Hemmer and A. Hansen, Journal of Applied Mechanics **59**, 909 (1992).
- [54] A. Hansen and P. C. Hemmer, Physics Letters A **184**, 394 (1994).
- [55] M. Kloster, A. Hansen, and P. C. Hemmer, Physical Review E **56**, 2615 (1997).
- [56] F. Kun, S. Zapperi, and H. J. Herrmann, The European Physical Journal B **17**, 269 (2000).
- [57] S. Pradhan, A. Hansen, and B. K. Chakrabarti, Reviews of Modern Physics **82**, 499 (2010).
- [58] R. C. Hidalgo, Y. Moreno, F. Kun, and H. J. Herrmann, Physical Review E **65**, 046148 (2002).
- [59] S. Pradhan, B. K. Chakrabarti, and A. Hansen, Physical Review E **71**, 036149 (2005).
- [60] S. Biswas and B. K. Chakrabarti, The European Physical Journal B **86**, 160 (2013).
- [61] Z. Danku, G. Ódor, and F. Kun, Phys. Rev. E. **98** (2018).
- [62] Y. Zhao, M. Chen, X. Jiang, X. Cao, and B. Qin, Developments in the Built Environment **21**, 100600 (2025).
- [63] T. Gutsman, G. E. Fantner, J. H. Kindt, M. Venturoni, S. Danielsen, and P. K. Hansma, Biophys. J. **86**, 3186 (2004).
- [64] R. B. Svensson, H. Mulder, V. Kovanen, and S. P. Magnusson, Biophys. J. **104**, 2476 (2013).
- [65] F. Jiang, H. Hörber, J. Howard, and D. J. Müller, J. Struct. Biol. **148**, 268 (2004).
- [66] J. Ruiz-Franco and J. van Der Gucht, Front. Cell Dev. Biol. **10**, 931776 (2022).
- [67] P. P. Provenzano and R. Vanderby, Jr, Matrix Biol. **25**, 71 (2006).
- [68] B. Innocenti and F. Galbusera, eds., *Human orthopaedic biomechanics* (Academic Press, San Diego, CA, 2022).